

References

- Hua Ai, Antoine Raux, Dan Bohus, Maxine Eskenazi, and Diane Litman. 2007. [Comparing spoken dialog corpora collected with recruited subjects versus real users](#). In *Proceedings SIGdial*.
- Frank J. Anscombe. 1954. [Fixed-sample-size analysis of sequential observations](#). *Biometrics*, 10:89–100.
- Matthias G. Arend and Thomas Schäfer. 2019. [Statistical power in two-level models: A tutorial based on Monte Carlo simulation](#). *Psychological methods*, 24(1):1–19.
- Erfan Sadeqi Azer, Daniel Khashabi, Ashish Sabharwal, and Dan Roth. 2020. [Not all claims are created equal: Choosing the right statistical approach to assess hypotheses](#). In *Proceedings of ACL*.
- Roy Bar-Haim, Ido Dagan, Bill Dolan, Lisa Ferro, Danilo Giampiccolo, Bernardo Magnini, and Idan Szpektor. 2006. [The second PASCAL recognising textual entailment challenge](#). In *Proceedings of the second PASCAL challenges workshop on recognising textual entailment*.
- Dale J. Barr, Roger Levy, Christoph Scheepers, and Harry J. Tily. 2013. [Random effects structure for confirmatory hypothesis testing: Keep it maximal](#). *Journal of Memory and Language*, 68(3):255–278.
- Colin B. Begg and Madhuchhanda Mazumdar. 1994. [Operating characteristics of a rank correlation test for publication bias](#). *Biometrics*, 50(4):1088–1101.
- Luisa Bentivogli, Peter Clark, Ido Dagan, and Danilo Giampiccolo. 2009. [The fifth PASCAL recognising textual entailment challenge](#). In *Proceedings of TAC*.
- Taylor Berg-Kirkpatrick, David Burkett, and Dan Klein. 2012. [An empirical investigation of statistical significance in NLP](#). In *Proceedings of EMNLP*.
- Katherine S. Button, John P. A. Ioannidis, Claire Mokrysz, Brian A. Nosek, Jonathan Flint, Emma S. J. Robinson, and Marcus R. Munafò. 2013. [Power failure: Why small sample size undermines the reliability of neuroscience](#). *Nature Reviews Neuroscience*, 14(5):365–376.
- Boxing Chen and Colin Cherry. 2014. [A systematic comparison of smoothing techniques for sentence-level BLEU](#). In *Proceedings of WMT*.
- Kevin Clark, Minh-Thang Luong, Urvashi Khandelwal, Christopher D. Manning, and Quoc V. Le. 2019a. [BAM! Born-again multi-task networks for natural language understanding](#). In *Proceedings of ACL*.
- Kevin Clark, Minh-Thang Luong, Quoc V. Le, and Christopher D. Manning. 2019b. [ELECTRA: Pre-training text encoders as discriminators rather than generators](#). In *Proceedings of ICLR*.
- Jacob Cohen. 1962. [The statistical power of abnormal-social psychological research: A review](#). *Journal of Abnormal and Social Psychology*, 65(3):145–153.
- John E. Connett, Judith A. Smith, and Richard B. McHugh. 1987. [Sample size and power for pair-matched case-control studies](#). *Statistics in Medicine*, 6(1):53–59.
- Ido Dagan, Oren Glickman, and Bernardo Magnini. 2005. [The PASCAL recognising textual entailment challenge](#). In *Proceedings of the Machine Learning Challenges Workshop*.
- Sumanth Dathathri, Andrea Madotto, Janice Lan, Jane Hung, Eric Frank, Piero Molino, Jason Yosinski, and Rosanne Liu. 2020. [Plug and play language models: A simple approach to controlled text generation](#). In *Proceedings of ICLR*.
- Janez Demšar. 2006. [Statistical comparisons of classifiers over multiple data sets](#). *J. Mach. Learn. Res.*, 7:1–30.
- Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2019. [BERT: Pre-training of deep bidirectional transformers for language understanding](#). In *Proceedings of NAACL*.
- Thomas G. Dietterich. 1998. [Approximate statistical tests for comparing supervised classification learning algorithms](#). *Neural computation*, 10(7):1895–1923.
- Jesse Dodge, Suchin Gururangan, Dallas Card, Roy Schwartz, and Noah A. Smith. 2019. [Show your work: Improved reporting of experimental results](#). In *Proceedings of EMNLP*.
- William B. Dolan and Chris Brockett. 2005. [Automatically constructing a corpus of sentential paraphrases](#). In *Proceedings of the Third International Workshop on Paraphrasing (IWP2005)*.
- Rotem Dror, Gili Baumer, Segev Shlomov, and Roi Reichart. 2018. [The hitchhiker’s guide to testing statistical significance in natural language processing](#). In *Proceedings of ACL*.
- Stephen W. Duffy. 1984. [Asymptotic and exact power for the McNemar test and its analogue with R controls per case](#). *Biometrics*, 40:1005–1015.
- Sergey Edunov, Myle Ott, Michael Auli, and David Grangier. 2018. [Understanding back-translation at scale](#). In *Proceedings of EMNLP*.
- Morten W. Fagerland, Stian Lydersen, and Petter Laake. 2013. [The McNemar test for binary matched-pairs data: mid-p and asymptotic are better than exact conditional](#). *BMC Medical Research Methodology*, 13.
- Cristina Garbacea, Samuel Carton, Shiyang Yan, and Qiaozhu Mei. 2019. [Judge the judges: A large-scale evaluation study of neural language models for on-line review generation](#). In *Proceedings of EMNLP*.

- Jonas Gehring, Michael Auli, David Grangier, Denis Yarats, and Yann N. Dauphin. 2017. [Convolutional sequence to sequence learning](#). In *Proceedings of ICML*.
- Andrew Gelman. 2019. [Don’t calculate post-hoc power using observed estimate of effect size](#). *Annals of Surgery*, 269(1):e9–e10.
- Andrew Gelman and John Carlin. 2014. [Beyond power calculations: Assessing type S \(sign\) and type M \(magnitude\) errors](#). *Perspectives on Psychological Science*, 9(6):641–651.
- Andrew Gelman and Jennifer Hill. 2006. *Data Analysis Using Regression and Multilevel/Hierarchical Models*. Cambridge University Press.
- Andrew Gelman and Eric Loken. 2013. [The garden of forking paths: Why multiple comparisons can be a problem, even when there is no “fishing expedition” or “p-hacking” and the research hypothesis was posited ahead of time](#).
- Danilo Giampiccolo, Bernardo Magnini, Ido Dagan, and Bill Dolan. 2007. [The third PASCAL recognizing textual entailment challenge](#). In *Proceedings of the ACL-PASCAL Workshop on Textual Entailment and Paraphrasing*.
- Yvette Graham, Nitika Mathur, and Timothy Baldwin. 2014. [Randomized significance tests in machine translation](#). In *Proceedings of WMT*.
- Tatsunori B Hashimoto, Hugh Zhang, and Percy Liang. 2019. [Unifying human and statistical evaluation for natural language generation](#). In *Proceedings of NAACL*.
- John M. Hoenig and Dennis M. Heisey. 2001. [The abuse of power: The pervasive fallacy of power calculations for data analysis](#). *The American Statistician*, 55(1):19–24.
- Ari Holtzman, Jan Buys, Maxwell Forbes, and Yejin Choi. 2020. [The curious case of neural text degeneration](#). In *Proceedings of ICLR*.
- John P. A. Ioannidis. 2019. [What have we \(not\) learnt from millions of scientific papers with \$P\$ values?](#) *The American Statistician*, 73(sup1):20–25.
- John P. A. Ioannidis, T. D. Stanley, and Hristos Doucouliagos. 2017. [The power of bias in economics research](#). *The Economic Journal*, 127(605):F236–F265.
- Shankar Iyer, Nikhil Dandekar, and Kornél Csernai. 2017. [First Quora dataset release: Question pairs](#).
- Philipp Koehn. 2004. [Statistical significance tests for machine translation evaluation](#). In *Proceedings of EMNLP*.
- Helena C. Kraemer and Christine Blasey. 2015. *How Many Subjects?: Statistical Power Analysis in Research*. SAGE.
- Peter A Lachenbruch. 1992. [On the sample size for studies based upon McNemar’s test](#). *Statistics in Medicine*, 11(11):1521–1525.
- Zhenzhong Lan, Mingda Chen, Sebastian Goodman, Kevin Gimpel, Piyush Sharma, and Radu Soricut. 2020. [ALBERT: A lite BERT for self-supervised learning of language representations](#). In *Proceedings of ICLR*.
- Chris van der Lee, Albert Gatt, Emiel van Miltenburg, Sander Wubben, and Emiel Krahmer. 2019. [Best practices for the human evaluation of automatically generated text](#). In *Proceedings of INLG*.
- Hector J. Levesque, Ernest Davis, and Leora Morgenstern. 2012. [The Winograd schema challenge](#). In *Proceedings of KR*.
- Yinhan Liu, Myle Ott, Naman Goyal, Jingfei Du, Mandar Joshi, Danqi Chen, Omer Levy, Mike Lewis, Luke Zettlemoyer, and Veselin Stoyanov. 2019. [ROBERTA: A robustly optimized BERT pre-training approach](#). *Computing Research Repository*, arXiv:1907.11692.
- R. Thomas McCoy, Junghyun Min, and Tal Linzen. 2019. [BERTs of a feather do not generalize together: Large variability in generalization across models with similar test set performance](#). *Computing Research Repository*, arXiv:1911.02969.
- Blakeley B. McShane, David Gal, Andrew Gelman, Christian Robert, and Jennifer L. Tackett. 2019. [Abandon statistical significance](#). *The American Statistician*, 73(sup1):235–245.
- Nathan Ng, Kyra Yee, Alexei Baevski, Myle Ott, Michael Auli, and Sergey Edunov. 2019. [Facebook FAIR’s WMT19 news translation task submission](#). In *Proceedings of WMT*.
- Daniel J. O’Keefe. 2007. [Brief report: Post hoc power, observed power, a priori power, retrospective power, prospective power, achieved power: Sorting out appropriate uses of statistical power analyses](#). *Communication Methods and Measures*, 1(4):291–299.
- Myle Ott, Sergey Edunov, Alexei Baevski, Angela Fan, Sam Gross, Nathan Ng, David Grangier, and Michael Auli. 2019. [FAIRSEQ: A fast, extensible toolkit for sequence modeling](#). In *Proceedings of NAACL*.
- Myle Ott, Sergey Edunov, David Grangier, and Michael Auli. 2018. [Scaling neural machine translation](#). In *Proceedings of WMT*.
- Kishore Papineni, Salim Roukos, Todd Ward, and Wei-Jing Zhu. 2002. [BLEU: A method for automatic evaluation of machine translation](#). In *Proceedings of ACL*.
- Jason Phang, Thibault Févry, and Samuel R Bowman. 2018. [Sentence encoders on STILTs: Supplementary training on intermediate labeled-data tasks](#). *Computing Research Repository*, arXiv:1811.01088.

- Matt Post. 2018. [A call for clarity in reporting BLEU scores](#). In *Proceedings of WMT*.
- Colin Raffel, Noam Shazeer, Adam Roberts, Katherine Lee, Sharan Narang, Michael Matena, Yanqi Zhou, Wei Li, and Peter J. Liu. 2019. [Exploring the limits of transfer learning with a unified text-to-text transformer](#). *Computing Research Repository*, arXiv:1910.10683.
- Pranav Rajpurkar, Robin Jia, and Percy Liang. 2018. [Know what you don't know: Unanswerable questions for SQuAD](#). In *Proceedings of ACL*.
- Pranav Rajpurkar, Jian Zhang, Konstantin Lopyrev, and Percy Liang. 2016. [SQuAD: 100,000+ questions for machine comprehension of text](#). In *Proceedings of EMNLP*.
- Stefan Riezler and John T. Maxwell. 2005. [On some pitfalls in automatic evaluation and significance testing for MT](#). In *Proceedings of the ACL Workshop on Intrinsic and Extrinsic Evaluation Measures for Machine Translation and/or Summarization*.
- Jeffrey D. Scargle. 1999. [Publication bias: The “file-drawer” problem in scientific inference](#). *arXiv*, arXiv:physics/9909033.
- James J. Schlesselman. 1982. *Case-control studies: Design, conduct, analysis*. Oxford University Press.
- Roy Schwartz, Jesse Dodge, Noah A. Smith, and Oren Etzioni. 2019. [Green AI](#). *Computing Research Repository*, arXiv:1907.10597.
- Richard Socher, Alex Perelygin, Jean Wu, Jason Chuang, Christopher D. Manning, Andrew Y. Ng, and Christopher Potts. 2013. [Recursive deep models for semantic compositionality over a sentiment treebank](#). In *Proceedings of EMNLP*.
- Anders Søgaard, Anders Johannsen, Barbara Plank, Dirk Hovy, and Hector Martínez Alonso. 2014. [What's in a \$p\$ -value in NLP?](#) In *Proceedings CoNLL*.
- Samy Suissa and Jonathan J. Shuster. 1991. [The 2 x 2 matched-pairs trial: Exact unconditional design and analysis](#). *Biometrics*, 47(2):361–372.
- Denes Szucs and John P. A. Ioannidis. 2017. [Empirical assessment of published effect sizes and power in the recent cognitive neuroscience and psychology literature](#). *PLoS biology*, 15(3).
- Eric-Jan Wagenmakers. 2007. [A practical solution to the pervasive problems of \$p\$ values](#). *Psychonomic Bulletin & Review*, 14:779–804.
- Alex Wang, Amanpreet Singh, Julian Michael, Felix Hill, Omer Levy, and Samuel R. Bowman. 2018. [GLUE: A multi-task benchmark and analysis platform for natural language understanding](#). In *Proceedings of the Workshop on BlackboxNLP*.
- Jacob Westfall, David A. Kenny, and Charles M. Judd. 2014. [Statistical power and optimal design in experiments in which samples of participants respond to samples of stimuli](#). *Journal of Experimental Psychology: General*, 143(5):2020–2045.
- Adina Williams, Nikita Nangia, and Samuel R. Bowman. 2018. [A broad-coverage challenge corpus for sentence understanding through inference](#). In *Proceedings of NAACL*.
- Zhilin Yang, Zihang Dai, Yiming Yang, Jaime Carbonell, Russ R. Salakhutdinov, and Quoc V. Le. 2019. [XLNet: Generalized autoregressive pretraining for language understanding](#). In *Proceedings of NeurIPS*.
- Georgios N. Yannakakis and Héctor P. Martínez. 2015. [Ratings are overrated!](#) *Frontiers in ICT*, 2.